EXOTHERMIC OR ENDOTHERMIC?

Name: ..... **Date:** .....

Exothermic reactions transfer energy to the surroundings (things get hotter, or light is given out). Endothermic reactions take energy in from the surroundings (things get colder, or energy must keep being supplied). Part 1: tick one. Part 2: complete the table. Part 3: full sentences.

PART 1: EXOTHERMIC OR ENDOTHERMIC?

1. A candle burning. **Exo** **Endo**
2. Photosynthesis in a leaf, which needs light energy. **Exo** **Endo**
3. A hand warmer that heats up when the iron inside it reacts with oxygen. **Exo** **Endo**
4. A sports cold pack that turns cold when you squeeze it. **Exo** **Endo**
5. Respiration in your cells, releasing energy from glucose. **Exo** **Endo**
6. Baking soda reacting with vinegar; the cup feels cooler. **Exo** **Endo**
7. Iron rusting, slowly releasing heat. **Exo** **Endo**
8. Splitting water into hydrogen and oxygen with electricity. **Exo** **Endo**
9. Neutralising an acid with an alkali; the thermometer reading rises. **Exo** **Endo**
10. Cooking an egg, which needs heat the whole time. **Exo** **Endo**

PART 2: OBSERVATIONS IN THE LAB

What you see	Evidence of a chemical reaction	Exothermic or endothermic?
Magnesium ribbon held in a Bunsen flame burns with a bright white light and leaves a white powder.		
Baking soda stirred into vinegar fizzes; the cup feels colder.		
An iron nail in blue copper sulfate solution: the solution turns pale green, a brown coating forms on the nail and the tube feels slightly warm.		
Hydrogen peroxide with a little yeast froths up, gives off oxygen and the beaker warms.		

PART 3: THE ENERGY STORY

In an exothermic reaction, where does the energy come from and where does it go?

Cooking an egg needs heat the whole time. Why does that make it endothermic?

EXOTHERMIC OR ENDOTHERMIC? ANSWER KEY

Name: ..... **Date:** .....

PART 1: EXOTHERMIC OR ENDOTHERMIC?

1. Exo 2. Endo 3. Exo 4. Endo 5. Exo 6. Endo 7. Exo 8. Endo 9. Exo 10. Endo

PART 2: OBSERVATIONS IN THE LAB

Magnesium | light and heat given out; a new white substance forms | exothermic

Baking soda and vinegar | gas produced (bubbles); temperature falls | endothermic

Iron in copper sulfate | colour change; a new solid (copper) forms; temperature rises | exothermic

Hydrogen peroxide and yeast | gas produced (froth); temperature rises | exothermic

PART 3: THE ENERGY STORY

1. Energy stored in the chemical bonds of the reactants is transferred to the surroundings as heat (and sometimes light), so the surroundings warm up. 2. The reaction takes in energy from the surroundings (the pan); if the heat supply stops, the reaction stops.

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